Maths Exploring for Exercises at Terminal-09

Progressive exercises

E1 · Direct application. In an orthonormous base of space, we consider $\vec{u} = (2; -1; 3)$ and $\vec{v} = (1; 4; 0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Application and justification. In an orthonormous basis, we consider $\vec{u} = (a; 2; 1)$ and $\vec{v} = (3; -1; 4)$where: a is real. a so that $\vec{u}$ and $\vec{v}$ be orthogonal.

E3 · Method to be organised. In an orthonormal coordinate system, a plan is directed by $\vec{u} = (1; -1; 0)$ and $\vec{v} = (1; 0; -1)$. Show that $\vec{n} = (1; 1; 1)$ He's normal. $A(0; 0; 0)$ and $B(2; 2; 2)$. Describe geometrically all the equidistant points of A and B.

E4 · Choice of strategy. In an orthonormous basis, we consider $\vec{u} = (1; 0; 1)$ and $\vec{v} = (0; 1; 1)$. Determine the measurement, between 0 and π, of the angle between these two vectors.

E5 · Synthesis or demonstration. A tetrahedron has as its base a right triangle with both sides
The height of the tetrahedron relative to this base is 5 cm.
Calculate volume in cm ^{3}.